# Demographic Covariates and Election Forensics: Re-evaluating Statistical Anomalies in South Korean Legislative and Presidential Elections (2020-2025)

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**Abstract**

This paper evaluates claims of artificial data generation in South Korean elections across four electoral cycles (2020-2025). Utilizing neighborhood-level precinct returns matched with demographic census records, we apply eleven statistical forensic tests alongside causal inference models. We find no evidence of systemic manipulation. The controversial 63:37 metropolitan early-voting ratio from 2020 is shown to be a cycle-specific artifact, with subsequent ratios dynamically tracking overall party performance. While nationwide digit tests pass cleanly in three of the four elections, we document an unresolved, candidate-specific last-digit anomaly in the 2022 Presidential Election ($\chi^2 = 24.98$, $p = 0.003$). Nevertheless, demographic covariates explain 49–73% of the early-voting gap across all four cycles. Ultimately, the statistical signatures of South Korea's elections mostly reflect age-based demographic sorting rather than top-down algorithmic fabrication.

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# 1 Introduction

The 21st South Korean legislative election, held in April 2020 amid the COVID-19 pandemic, saw record turnout driven largely by an unprecedented surge in early voting. Following the election, two interrelated anomalies attracted public attention. First, the ruling Democratic Party consistently outperformed in early voting compared to same-day voting by approximately 9–10 percentage points. Second, the two-party early-voting ratio across the Seoul, Incheon, and Gyeonggi metropolitan corridor appeared to converge to approximately 63:37 in favour of the Democrats, a regularity critics argued was too precise to arise naturally. These patterns led to public allegations of systemic irregularities and artificial data generation.

Election forensics - a subfield of political methodology - utilises statistical tools to detect artificial manipulation in vote counts (Beber and Scacco, 2012). In the wake of the 2020 election, an initial working paper utilising the *eforensics* statistical model flagged the South Korean early voting data as highly anomalous (Mebane, 2020). However, flawed forensic research can have profound consequences; Kuk et al. (2024) demonstrate that publishing methodologically deficient election forensics severely erodes democratic trust and intensifies political polarisation.

Forensic models are critically vulnerable to omitted variable bias. Domestic scholars noted that initial anomaly reports failed to account for unique domestic data formatting (Park, 2020). Yoo and Min (2021) demonstrated that when age demographics and regional sorting are properly modelled, the statistical probability of artificial manipulation approaches zero.

A second vulnerability - equally important for the present study - concerns the naive application of the Second-Digit Benford's Law (2BL) test. Shikano and Mack (2011) apply the 2BL test to the 2009 German Federal Parliamentary Election, against which no fraud allegation has been raised, and find that the standard $\chi^2(9)$ distribution is inappropriate as a reference distribution when two conditions are jointly met at the station level: the mode of the vote-count distribution lies between approximately 80 and 190, and the distribution is homogeneous (KDE maximum $> 0.01$). As we show

below, these station-level thresholds do not apply when the analysis unit is the Dong-level aggregate rather than the individual polling booth; nevertheless, the Shikano and Mack (2011) framework provides essential context for interpreting marginal 2BL results in geographically restricted subsets.

The present paper extends analysis to two additional electoral cycles: the 20th Presidential Election (March 2022) and the 21st Presidential Election (June 2025). The 2022 presidential, decided by a margin of less than one percentage point, attracted significant controversy over the early-voting gap, while the 2025 snap presidential election - held following the impeachment of President Yoon Suk-yeol - provides a high-stakes out-of-sample test under dramatically different political conditions and a large opposition majority. Together, the four elections spanning 2020–2025 form a natural quasi-experimental sequence in which the Democratic Party's overall performance varied substantially (from a legislative landslide in 2020 to a razor-thin presidential loss in 2022, a legislative majority in 2024, and a decisive presidential victory in 2025), enabling robust cross-election inference about the structural origins of early-voting anomalies.

## 2 Data and Methodology

We construct microscopic datasets linking precinct-level election returns from the National Election Commission (NEC) to neighbourhood-level (Eup/Myeon/Dong) demographic data from the Korean Statistical Information Service (KOSIS), covering all four electoral cycles. The 21st General Election dataset includes $N = 3{,}479$ elementary districts; the 20th Presidential Election $N = 3{,}509$; the 22nd General Election $N = 2{,}726$; and the 21st Presidential Election $N = 3{,}553$. All datasets are restricted to Democratic candidate early vote counts of at least ten and early/same-day totals above 50. We do not deal with proportional representative ('biryedaepyo') votes in legislative elections, focusing only on regional representatives.

## 2.1 Data Sources and Construction

Three primary administrative datasets are merged to construct the analytical files.

**Election results.** Precinct-level vote counts are drawn from the National Election Commission open data portal. For legislative elections, the source is the Korean open data portal at `https://www.data.go.kr/data/15025527/fileData.do`; for presidential elections, the corresponding dataset is at `https://www.data.go.kr/data/15025528/fileData.do`. Each CSV file records votes at the individual polling-station level (*tupyogumyeong*), disaggregated by candidate and voting method. The NEC distinguishes three voting types: same-day station votes, in-precinct early votes (*gwannaesajeoninpyo*; code variable `in_precinct_early`), and out-of-precinct absentee early votes (*gwanoesajeoninpyo*; code variable `out_precinct_early`). A critical feature of the raw data is that the NEC records registered voters (*seongeoinsu*) separately for each voting-method row: same-day station rows report only the registered voters who did *not* vote early, while the in-precinct early row records those who did. Summing only the same-day rows therefore systematically undercounts total registered voters by approximately 37% (the early-voting participation rate), producing computed turnout rates above 100% in many Dongs. The pipeline corrects for this by summing both row types to recover the true total electoral roll, and applies the analogous correction for out-of-precinct absentee voters at the constituency level. Presidential election files use slightly different column names (*sigungumyeong* for constituency, *eupmeondongmyeong* for Dong), which the pipeline normalises to the internal schema.

**Demographic census data.** Monthly age–sex resident registration counts at the Dong level are sourced from Korea's Ministry of the Interior and Safety (MOIS) resident registration statistics portal at `https://jumin.mois.go.kr/ageStatMonth.do`, accessible via the Korean open data portal at `https://www.data.go.kr/data/3033304/fileData.do`. Each CSV file provides single-year-of-age population counts disaggregated by sex for all administrative units at a given month-year snapshot. The pipeline aggregates these into 22 five-year age–sex cohorts (ages 18–24 through 70+, separately for males and females), expressed as proportions of the total voting-age population. A com-

posite demographic propensity score $W_i$ - the proportion of 40–59-year-olds in the total voting-age population - is also constructed as a mobilisation weight, reflecting the well-documented propensity of middle-aged working voters to disproportionately use early voting. Because presidential election CSV files do not include a separate census file, the pipeline falls back on the nearest available legislative-election census vintage (22nd General census for the 21st Presidential; 21st General census for the 20th Presidential). Dong-to-census matching is performed via a multi-pass normalisation algorithm that strips administrative suffixes, applies fuzzy string matching at a similarity cutoff of 0.82, and cascades through the candidate constituency keys.

**Apartment price data.** Monthly apartment transaction records are sourced from Korea's Ministry of Land, Infrastructure and Transport (MOLIT) real estate transaction disclosure portal at `https://rt.molit.go.kr/pt/xls/xls.do?mobileAt=`, accessible via the Korean open data portal at `https://www.data.go.kr/data/3050988/fileData.do`. Each record contains the transaction price (in units of 10,000 Korean Won), exclusive floor area ($m^2$), and the administrative address. The pipeline computes a price-per-square-metre figure for each transaction, then collapses to the median at the Dong level as a stable measure of neighbourhood wealth. Missing Dong-level medians are filled sequentially from the constituency-level median and then the province-level median. The log-transformed median price per square metre (`log_apt_price`) enters all regression models as a continuous wealth proxy. Presidential election files inherit the apartment price data from the nearest legislative election vintage using the same fallback mechanism as the census. We use the twelve-month pre-election apartment price data for each election.

All three datasets are matched at the Dong level using a common administrative key constructed from normalised province, constituency, and Dong name strings. The final analytical Dong-level file retains only observations with non-missing demographic propensity and at least one female 40–44 cohort observation (`pct_f_4044`), which serves as a completeness filter for the census merge.

## 2.2 Bayesian vs. Frequentist Frameworks in Election Forensics

The *eforensics* model of Mebane (2020), as well as the econometric counter-analysis of Yoo and Min (2021), rely on Bayesian inference. Specifically, *eforensics* employs a finite mixture model that classifies voting units into unobserved latent states (e.g., 'no fraud', 'incremental fraud', 'extreme fraud') by estimating posterior probability distributions. Such models are exceptionally vulnerable to omitted variable bias: if the 'no fraud' baseline distribution is specified without incorporating critical covariates - such as the highly polarised age demographics that dictated early-voting usage across all four electoral cycles - the model will force the resulting natural divergence into the 'fraud' classification.

Our frequentist approach directly tests the observed data against mathematically established null hypotheses, requiring far fewer assumptions about the underlying probability distribution of voting behaviour. This approach is arguably superior in highly polarised environments precisely because it avoids parameterising an unobserved 'normal' distribution.

## 2.3 The Interpretation of $P$-Values in Population Data

Because election returns constitute the entire population of voting records rather than a sample, the $p$-value shifts from measuring sampling error to evaluating the *data-generating mechanism* (Fisher, 1925). The null hypothesis is that vote counts are the product of a natural, uncoordinated stochastic process - millions of independent human decisions interacting with normal geographic variance. A $p$-value below $\alpha = 0.05$ indicates incompatibility with this natural generating process. Conversely, a nominally significant $p$-value driven solely by a large sample size carries no forensic meaning; the appropriate diagnostic in such cases is visual inspection, such as the Q-Q plot.

## 2.4 Methodological Limitations of the Naive 2BL $\chi^2$ Test

Shikano and Mack (2011) demonstrate that the standard application of the 2BL test - comparing the $\chi^2$ statistic against the $\chi^2(9)$ distribution with critical value 16.9 at

$\alpha = 5\%$ - produces systematically inflated test statistics under two empirically common conditions: (1) the mode of the vote-count distribution lies between approximately 80 and 190, and (2) the distribution is homogeneous, operationalised as a kernel density estimate (KDE) maximum above approximately 0.01. They demonstrate this using simulation-calibrated critical values on the 2009 German Federal Election, where the naive test falsely flags many constituencies as anomalous while the simulation-based test restores expected failure rates.

A critical qualification governs the application of the Shikano and Mack (2011) framework to the present study. Their thresholds - mode 80–190 and KDE max $> 0.01$ - were calibrated on individual polling-station (*Wahlbezirk*) vote counts in Germany, which typically range from approximately 50 to 500 votes per station. Our [F7a] analysis operates on Dong-level *aggregates*: each Dong combines the results of multiple polling booths into a single observation, producing vote totals in the thousands for legislative elections. As confirmed by the diagnostic output for the legislative elections, Dong-level modes of approximately 1,986 and 2,799 votes and KDE maxima far below 0.01 place these datasets entirely outside the S&M calibration range.

The situation is more nuanced for the 20th Presidential Election (2022), where the metro Dong-level mode of approximately 2,278 votes lies well outside the S&M calibration range. For the 21st Presidential Election (2025), the metro Dong-level mode of approximately 2,750 votes again places the data clearly outside the S&M station-level parameters.

Additionally, as Fewster (2009) establishes, Benford's Law is most reliable when the underlying data span several orders of magnitude on the original scale. Across all four elections, the nationwide distributions span 2.64–2.75 orders of magnitude, comfortably in the multi-order-of-magnitude regime. The presidential elections, covering all precincts without legislative-district filtering, span a wider absolute range than the legislative elections.

## 2.5 Standard Forensic Metrics [F1]–[F12]

We apply eleven test groups:

1. **[F1] Second-Digit Benford's Law (2BL), nationwide:** Applied to the full nationwide Dong-level in-precinct early vote totals (`in_precinct_early_dem`). The expected probability of second digit $d$ is $P(d) = \sum_{k=1}^{9} \log_{10}(1+1/(10k+d))$ (Mebane, 2011). The nationwide test is authoritative because the nationwide distribution spans the full range of precinct sizes. We use the naive $\chi^2(9)$ test and note the limitation of Section 2.4; the nationwide distribution's high heterogeneity (spanning rural precincts of tens of votes to large urban ones of thousands) makes the $\chi^2$ approximation more reliable here than in restricted subsets.

2. **[F2] Last-Digit Uniformity, nationwide:** Terminal digits of sufficiently large vote counts should follow a uniform distribution; deviations indicate algorithmic padding or human rounding (Beber and Scacco, 2012). Unlike the 2BL, the last-digit test is substantially less sensitive to distributional homogeneity effects, making it a useful corroborating or contrasting test when 2BL results are ambiguous. The test is applied separately to three vote series - in-precinct early Democratic votes (`in_precinct_early_dem`), in-precinct early conservative votes (`in_precinct_early_con`), and same-day Democratic votes (`same_day_dem`) - enabling detection of asymmetric failures that could distinguish candidate-specific anomalies from system-wide artefacts. A supplementary binomial simulation null is also computed, drawing from the empirical $(N_i, p_i)$ pair at each Dong to establish the expected last-digit distribution under pure random sampling; this provides a sharper null than the flat uniform assumption.

3. **[F3] Variance Dispersion:** Mean within-city standard deviation of Democratic vote shares across neighbourhoods; tests for artificial clustering toward a top-down target.

4. **[F4] Invalid Vote Correlation:** Pearson correlation between a station's invalid vote rate and its Democratic vote share across all individual station observations.

5. **[F5]/[F6] Regional Aggregate Analysis:** Macroscopic Democratic two-party vote ratios across the Seoul, Incheon, and Gyeonggi zones, and micro-level station standard deviation; tests whether macro convergence is consistent with the Law of Large Numbers.

6. **[F7] Metro Early-Voting Analysis (63:37 Hypothesis):** Restricted to Seoul, Incheon, and Gyeonggi precincts, in-precinct early votes only. Comprises: [F7a] 2BL with the Shikano and Mack (2011) homogeneity diagnostic and naive $\chi^2(9)$ result; [F7b] last-digit uniformity on the same metro subset; [F7c] early-only two-party ratio per metropolitan zone with micro-level standard deviation; and [F7d] simulation-adjusted 2BL per constituency, implemented via the Shikano and Mack (2011) procedure - a multivariate Student-$t$ is fitted to the Dong-level log-ratio of vote shares within each constituency, 500 synthetic elections are drawn, and the 95th-percentile chi-square value (chi2_sim95) provides a constituency-specific critical value; the empirical chi-square is then scaled to the standard reference distribution (chi2_trans). The share of metro constituencies with chi2_trans $> 16.9$ constitutes the rejection rate; under the null this should approximate 5%, and a substantially higher rate constitutes evidence of systematic digit anomalies across the metro subset.

7. **[F8] Beber and Scacco '0s and 5s' Test:** Human fabricators display a systematic preference for round digits 0 and 5 when inventing numbers. Under the null hypothesis of natural digit generation, the combined frequency of terminal digits 0 and 5 should be approximately 20%. A statistically significant excess constitutes evidence of human fabrication (Beber and Scacco, 2012).

8. **[F9] Digit Mean and Variance Test:** Under a uniform distribution, the expected mean of the last digit is 4.5 and the expected variance is 8.25. Significant deviation from either moment indicates non-random digit generation.

9. **[F10] Klimek Mixture Model (Election Fingerprint):** Klimek et al. (2012) demonstrate that fraudulent elections characteristically produce a cluster of precincts

exhibiting both abnormally high turnout and abnormally high vote shares for the ruling party *simultaneously* - the 'election fingerprint'. We fit a Gaussian mixture model to the joint distribution of precinct-level turnout and Democratic vote share, with the number of components selected by Bayesian Information Criterion (ranging from 2 to 6). Each cluster is classified according to the Klimek joint criterion: a cluster is flagged as anomalous only when its centroid simultaneously exceeds 90% turnout *and* 90% (or below 10%) vote share. A cluster that is extreme only in vote share at normal turnout is classified as geographic or partisan, not fraudulent. The dominant province of each cluster is further checked against known strongly partisan regions (Honam: Jeonnam, Jeonbuk, Gwangju; Yeongnam: Gyeongbuk, Gyeongnam, Daegu, Busan, Ulsan) to prevent mislabelling structural partisanship as fraud.

10. **[F11] Sebastopol Test (Turnout-Share Correlation):** Myagkov et al. (2009) documented a positive within-constituency correlation between precinct-level turnout and vote share as a mechanical signature of ballot stuffing in post-Soviet Russian and Ukrainian elections, identifying the pattern with particular clarity in data from the city of Sebastopol (Sevastopol); the test bears that city's name. Within each constituency, we compute the Pearson correlation between Dong-level turnout and Democratic vote share. A correlation above 0.70 within a constituency is a potential signature of ballot stuffing, in which turnout inflation and vote-share inflation co-vary within a geographic unit. Constituencies flagging in multiple elections, or with small Dong counts and known structural demographic features, receive additional interpretive scrutiny before a manipulation conclusion is drawn.

11. **[F12] Absentee Shift Analysis:** A comparative evaluation of the gap between out-of-precinct early (absentee) Democratic vote share and same-day Democratic vote share, contrasted against the standard in-precinct early gap. Evaluated via a constituency-level one-sample t-test, this metric identifies whether the absentee voting pool systematically diverges in partisan composition from the local early-voting pool across different election types.

### 2.6 Demographic Weighting, WLS Regression, and PSM

Middle-aged voters (40–59 years old) exhibit a significantly higher propensity for early voting due to weekday employment constraints, while displaying distinct partisan preferences. This creates a systematic early-voting selection bias that inflates the raw gap in areas with dense middle-aged populations. We correct for this by computing a demographic concentration score $W_i$ (the ratio of the middle-aged cohort to the total voting-age population) and evaluating the adjusted gap $G_{\text{adj},i} = G_{\text{raw},i}/W_i$ alongside the raw gap.

The WLS mobilisation model [C1] regresses the raw gap on all 21 five-year age-gender cohorts (excluding one as reference), urbanisation type (`urban_type`: Dong, Eup, or Myeon, derived from the administrative suffix of each unit's name), log apartment price per square metre, region fixed effects (`province_tag`), and military zone indicators (`is_military`). A high $R^2$ is evidence for organic demographic mobilisation. A separate WLS model [C2] regresses in-precinct early Democratic vote share (`early_pct`) on same-day share plus all covariates; a strict algorithmic multiplier would force $R^2 \to 1.0$. Propensity Score Matching (PSM) [C3] (Rosenbaum and Rubin, 1983) matches above-median early-turnout precincts to controls on the same 21 covariates. A supplementary WLS Variance Analysis [C4] reports the unadjusted proportion of gap variance explained using the same covariate set. All continuous covariates are standardised prior to regression using z-score normalisation. Standard errors are heteroskedasticity-robust (HC3).

## 3 Results

Figures 1–4 show the full forensic dashboards for all four elections. Code is available at the repository in the appendix.

### 3.1 21st General Election (2020): Nationwide Digit Tests [F1] and [F2]

**2BL [F1]:** Applied to the full $N = 3{,}479$ Dong-level `in_precinct_early_dem` totals, the chi-square test yields $\chi^2 = 9.36$ ($p = 0.404$, $df = 9$). We fail to reject the null; second

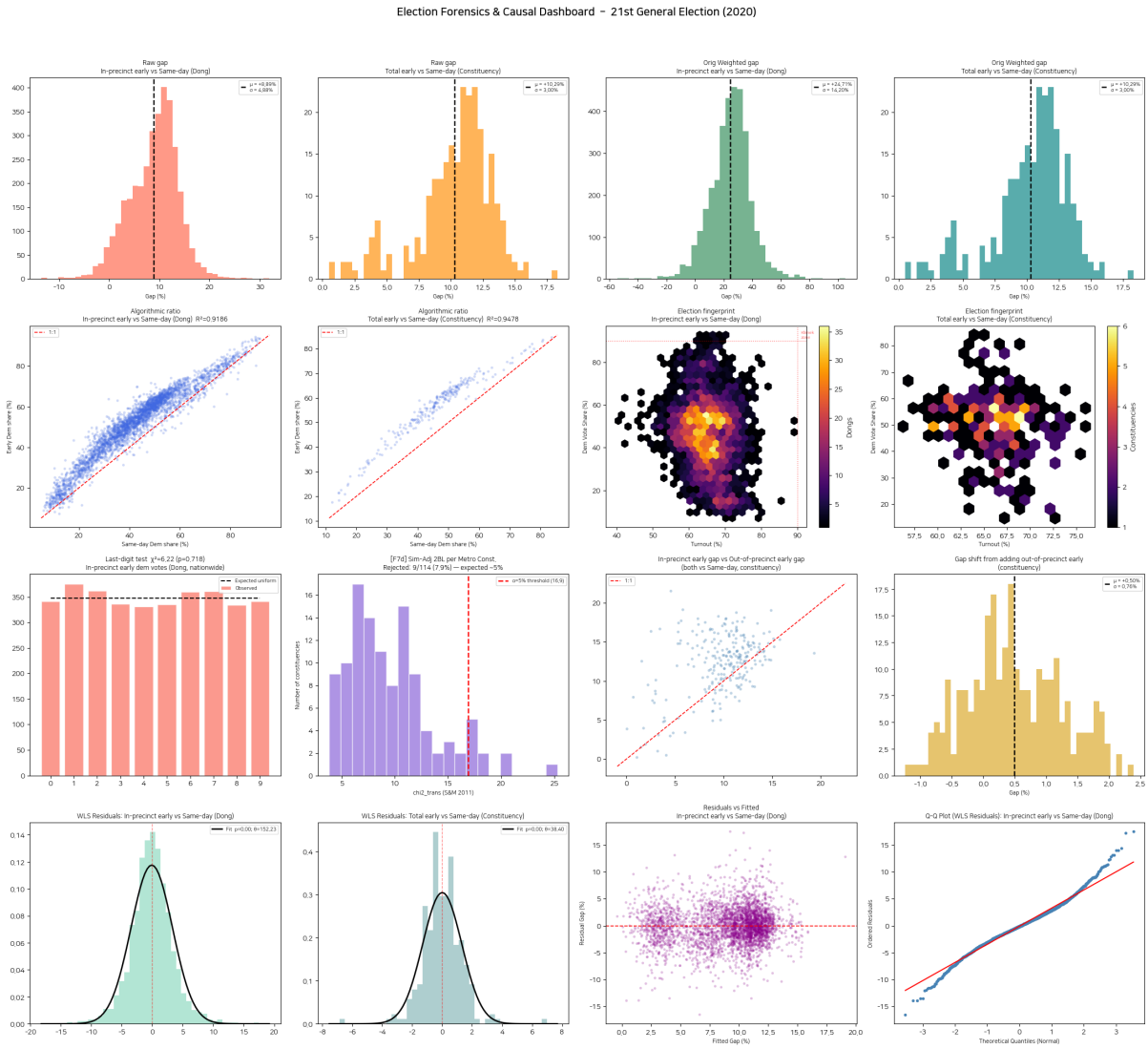


Figure 1: Election forensics and causal dashboard, 21st General Election (2020). Row 3 (left): last-digit uniformity [F2]; row 3 (right): Second-Digit Benford [F1]. Row 4: WLS residual histogram, WLS constituency residuals, residuals-vs-fitted, and Q-Q plot.

digits conform to the natural Benford distribution. The nationwide distribution spans 24–11,483 votes (2.68 orders of magnitude), placing it well within the heterogeneous regime where the Shikano and Mack (2011) homogeneity conditions are unlikely to be met.

**Last-Digit Uniformity [F2]:** Applied to three independent series: `in_precinct_early_dem` ($\chi^2 = 6.22$, $p = 0.718$), `in_precinct_early_con` ($\chi^2 = 9.59$, $p = 0.385$), and `same_day_dem` ($\chi^2 = 9.53$, $p = 0.390$). All three pass comfortably. Terminal digits are uniformly distributed across all series; no algorithmic padding signature.

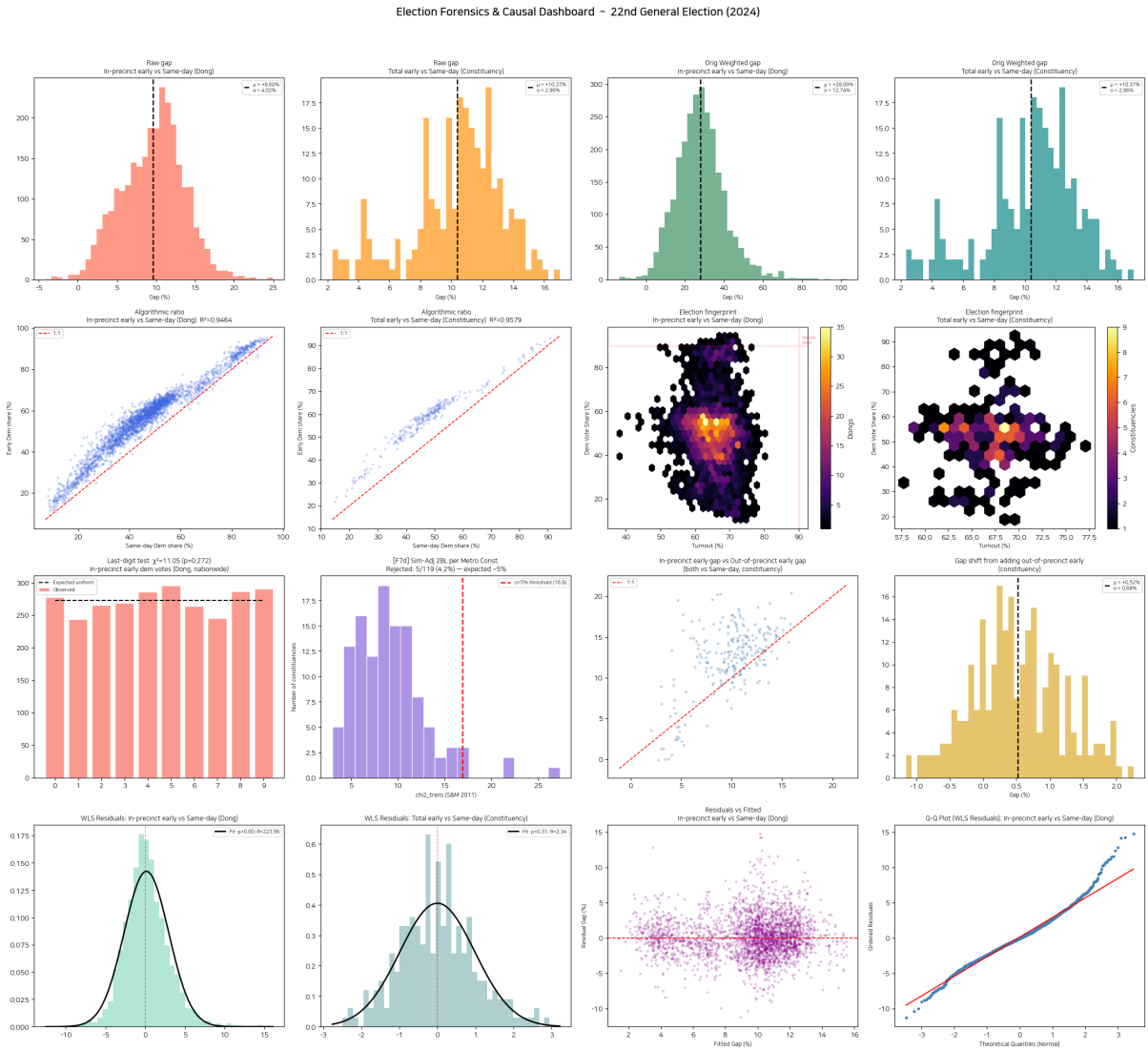


Figure 2: Election forensics and causal dashboard, 22nd General Election (2024). Layout identical to Figure 1.

## 3.2 21st General Election (2020): Variance Dispersion [F3], Covariate Standardisation, and PSM

The mean within-city standard deviation of Democratic vote shares is 5.85%, confirming natural geographic dispersion.

The raw mean early-to-same-day gap is +8.89% ($\sigma$ = 4.88%) at Dong level and +10.29% ($\sigma$ = 3.00%) at constituency level. This high baseline variance already disproves uniform algorithmic padding. The WLS mobilisation model [C1] explains 49.0% of gap variance (adj. $R^2$ = 0.484; $F$ = 89.67; $N$ = 3,479). The bivariate $r^2$ between `same_day_dem` share and `early_pct` is 0.9186 (Dong) and 0.9478 (constituency); the full

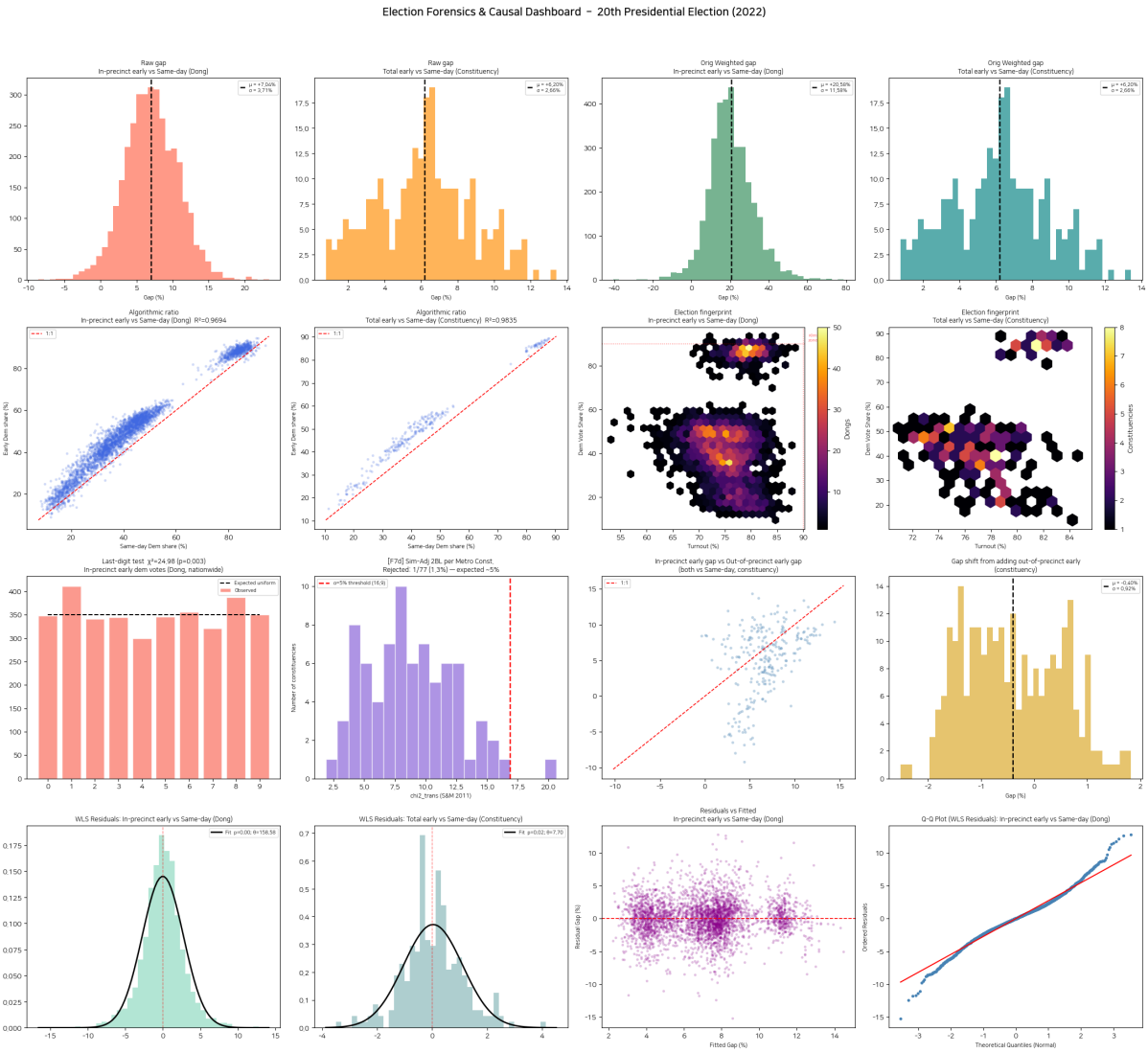


Figure 3: Election forensics and causal dashboard, 20th Presidential Election (2022). Layout identical to Figure 1. Note the elevated digit-5 bar in the last-digit panel (row 3, left), corresponding to the nationwide last-digit failure [F2] ($\chi^2 = 24.98$, $p = 0.003$) confined to the `in_precinct_early_dem` series.

WLS [C2] model achieves $R^2 = 0.961$. The residual variance in both specifications refutes a strict algorithmic multiplier.

The Q-Q plot and residual histogram (Figure 1, bottom row) of the WLS residuals confirm approximate normality in the central mass. PSM (threshold: $> 2{,}247$ early votes) yields ATT $= +4.53$ pp, consistent with the partisan geography of high-density, middle-aged urban precincts.

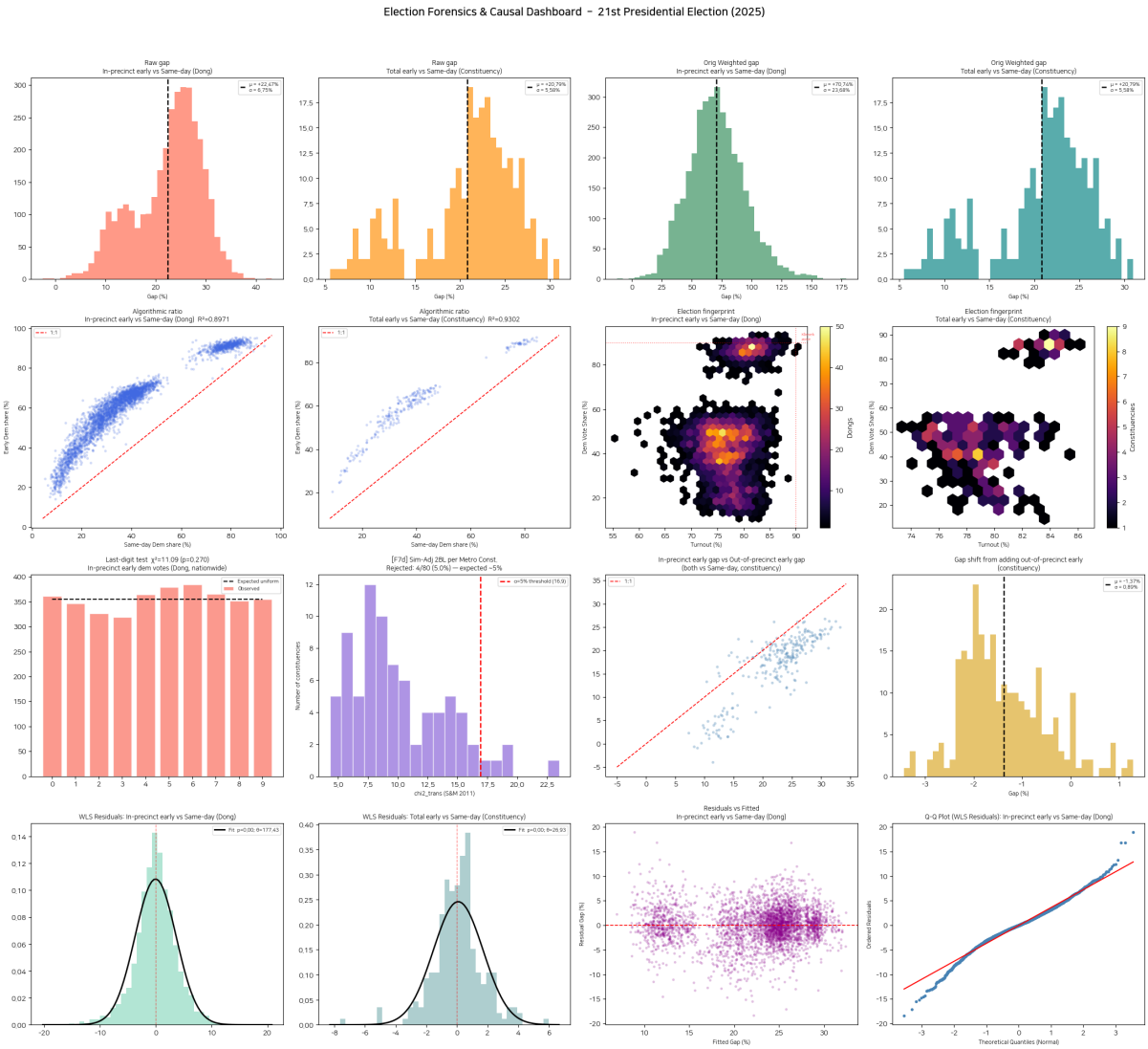


Figure 4: Election forensics and causal dashboard, 21st Presidential Election (2025). Layout identical to Figure 1. Note the substantially larger raw gap distributions in row 1 ($\mu \approx +22\%$), consistent with the Democratic candidate's landslide victory.

## 3.3 21st General Election (2020): Regional Aggregates [F5/F6] and Invalid Votes [F4]

**Overall two-party ratios [F6]:** Democratic shares: Gyeonggi 55.94% ($n$ = 3,731), Incheon 56.80% ($n$ = 871), Seoul 54.99% ($n$ = 2,676). Micro standard deviation across all metropolitan stations: 11.06%.

**Invalid vote correlation [F4]:** $r = -0.039$ across $N = 17{,}799$ stations. The near-zero and negative effect size confirms that invalid votes were independent statistical noise, not weaponised to offset fabricated totals.

## 3.4 21st General Election (2020): Metro Early-Voting Analysis [F7]

**Metro 2BL [F7a]:** For $N = 1{,}122$ metro precincts, $\chi^2 = 16.38$ ($p = 0.059$, $df = 9$) - a borderline pass, marginally above the critical value of 16.9. The scale-range comparison shows the nationwide range spans 24–11,483 votes (2.68 orders) while the metro range spans 41–11,483 votes (2.45 orders); both are comfortably multi-order-of-magnitude, with the metro subset differing only in the lower tail where small rural precincts are absent. The Shikano and Mack (2011) diagnostic confirms that neither S&M condition is met at Dong-level aggregation (metro mode $\approx$ 1,986 votes; KDE max $\approx$ 0.00036). The borderline result cannot therefore be explained by the S&M station-level inflation mechanism. The mild elevation relative to the nationwide $\chi^2 = 9.36$ is consistent with the narrower range of partisan environments in the metro Dong subset compared with the full nationwide distribution, but at $p = 0.059$ the result is a borderline pass that warrants no forensic weight on its own.

**Simulation-adjusted metro 2BL [F7d]:** 9 of 114 constituencies (7.9%) reject at the 5% threshold, against an expected rejection rate of approximately 5% under the null. This is slightly above the expected rate, consistent with the borderline naive [F7a] result, but the excess is small (9 vs. an expected $\approx 5.7$) and a one-sided binomial probability of $P(X \geq 9 \mid n = 114, p = 0.05) \approx 0.27$ indicates no statistically significant accumulation. Taken with the borderline native pass, the [F7d] result provides no forensically actionable signal.

**Metro last-digit [F7b]:** $\chi^2 = 13.28$ ($p = 0.150$, $df = 9$). A comfortable pass on the same metro precincts, directly contradicting any hypothesis of digit-level algorithmic padding.

**Early-only two-party ratio [F7c]:** The macro in-precinct early-voting two-party Democratic shares are: Gyeonggi 63.27% vs 36.73% ($n = 545$), Incheon 63.07% vs 36.93% ($n = 154$), Seoul 63.48% vs 36.52% ($n = 424$). The 63:37 claim is numerically confirmed for the 21st election. The micro-level standard deviation of individual early-station two-party shares is 9.14%. This high variance is incompatible with any top-down constraint:

a fixed algorithmic target would require near-zero micro-level variance. The macro convergence is instead the natural product of the Law of Large Numbers aggregating highly dispersed local outcomes.

## 3.5 21st General Election (2020): Additional Forensic Tests [F8]–[F11]

**[F8] Beber and Scacco '0s and 5s':** Observed frequency 19.46% ($Z = -0.797$, $p = 0.426$) - a clear pass, consistent with no human fabrication signature.

**[F9] Digit mean and variance:** Mean = 4.468 (expected: 4.500); variance = 8.280 (expected: 8.250). Both moments consistent with the uniform distribution.

**[F10] Klimek mixture model:** The BIC-selected four-component model (BIC = −13,250.2) identifies a swing/mixed-region cluster (39.1% of precincts; mean turnout 64.3%, Democratic share 54.1%), a Yeongnam-dominant cluster (31.4%; 65.5%, 39.9%), a deep-conservative Yeongnam cluster (17.5%; 67.7%, 20.7%), and a Honam cluster (12.0%; 64.5%, 73.4%). None meets the Klimek joint fraud criterion; cluster separation is explained entirely by known regional partisanship.

**[F11] Sebastopol test:** One constituency flags: Sejong ($r = 0.795$, $n = 19$ Dongs). Sejong Metropolitan Autonomous City is a special administrative unit with an atypically small number of Dongs and a rapidly growing, demographically homogeneous population. As documented below, Sejong flags in all four elections examined, strongly suggesting a structural feature of the constituency rather than cycle-specific manipulation.

## 3.6 22nd General Election (2024): Nationwide Digit Tests [F1] and [F2]

**2BL [F1]:** $\chi^2 = 10.51$ ($p = 0.311$, $df = 9$). Nationwide second digits of `in_precinct_early_dem` conform to the Benford distribution.

**Last-Digit Uniformity [F2]:** Applied to three series: `in_precinct_early_dem` ($\chi^2 = 11.05$, $p = 0.272$), `in_precinct_early_con` ($\chi^2 = 6.65$, $p = 0.674$), and `same_day_dem`

($\chi^2 = 7.43$, $p = 0.593$). All three pass; terminal digits are uniformly distributed.

## 3.7 22nd General Election (2024): Variance Dispersion [F3], Covariate Standardisation, and PSM

Mean within-city vote-share standard deviation: 5.25%.

The raw early-to-same-day gap is +9.60% ($\sigma = 5.02\%$) at Dong level and +10.37% ($\sigma = 2.96\%$) at constituency level. WLS [C1]: adj. $R^2 = 0.507$, $F = 95.83$, $N = 2{,}726$. Bivariate $r^2$: 0.9464 (Dong), 0.9579 (constituency). WLS [C2]: $R^2 = 0.968$. The Q-Q plot and residual histogram of the WLS model confirm approximate normality. PSM ATT = +2.09 pp - a small positive value reflecting natural demographic differences between matched precincts.

## 3.8 22nd General Election (2024): Regional Aggregates [F5/F6] and Invalid Votes [F4]

**Overall two-party ratios [F6]:** Gyeonggi 55.18% ($n = 3{,}828$), Incheon 53.47% ($n = 873$), Seoul 51.71% ($n = 2{,}682$). Micro standard deviation: 9.52%.

**Invalid vote correlation [F4]:** $r = +0.081$ across $N = 16{,}482$ stations - negligible, and with opposing sign to 2020.

## 3.9 22nd General Election (2024): Metro Early-Voting Analysis [F7]

**Metro 2BL [F7a]:** For $N = 1{,}124$ metro precincts, $\chi^2 = 18.00$ ($p = 0.035$, $df = 9$) - a marginal failure, just above the critical value of 16.9. Five considerations jointly argue against a fraud interpretation.

*First, the S&M diagnostic rules out station-level inflation.* The Shikano and Mack (2011) diagnostic confirms that neither S&M condition is met (metro mode $\approx$ 2,799 votes; KDE max $\approx$ 0.00028). Their inflation mechanism - which operates when station-level vote counts cluster in the 80–190 range - does not apply to Dong-level aggregates in the

thousands. The naive $\chi^2$ test is therefore reliable at this aggregation level, and the failure is a genuine statistical result.

*Second, cross-election consistency.* Both legislative elections produce nearly identical metro 2BL statistics: $\chi^2 = 16.38$ (21st, borderline pass) and $\chi^2 = 18.00$ (22nd, marginal fail), a difference of only $\Delta\chi^2 = 1.62$. If the 22nd failure reflected genuine manipulation, one would expect a substantially larger statistic (e.g., $\chi^2 > 40$, as documented in fraudulent settings in the literature) and a much smaller $p$-value. The near-identical values across elections point instead to a stable structural feature of metro Dong-level data.

*Third, the nationwide test passes cleanly.* The full $N = 2{,}726$ nationwide 2BL yields $\chi^2 = 10.51$ ($p = 0.311$). Selective manipulation confined to metro precincts while leaving nationwide digit properties intact would require an implausibly narrow targeting assumption.

*Fourth, the magnitude is trivially above threshold.* The critical value for $\alpha = 5\%$ under $\chi^2(9)$ is 16.9; the observed statistic of 18.00 exceeds this by 1.10. Forensically significant manipulation in election studies typically produces chi-squares many times the critical value, not a result that falls in the same order of magnitude as the threshold.

*Fifth, the simulation-adjusted constituency-level test passes.* When the 2BL is applied per constituency to the metropolitan subset, only 5 of 119 constituencies (4.2%) reject at the 5% threshold - below the expected rejection rate of 5% under the null. This directly contradicts any systematic pattern of constituency-level digit anomalies.

**Metro last-digit [F7b]:** $\chi^2 = 12.17$ ($p = 0.204$, $df = 9$). Passes on the same metro data, directly contradicting digit-level manipulation.

**Early-only two-party ratio [F7c]:** The macro early-voting Democratic shares are: Gyeonggi 61.78% vs 38.22% ($n = 565$), Incheon 59.02% vs 40.98% ($n = 135$), Seoul 59.61% vs 40.39% ($n = 425$). The 63:37 ratio does *not* replicate in 2024. All three zones fall below their 2020 levels by 1–4 percentage points, tracking the Democratic Party's lower overall vote share in 2024. This dynamic behaviour - ratios tracking overall Democratic performance rather than converging to a fixed value - is incompatible with any fixed algorithmic target. The micro-level standard deviation is 7.04%.

## 3.10 22nd General Election (2024): Additional Forensic Tests [F8]–[F11]

**[F8] Beber and Scacco '0s and 5s':** Observed frequency 21.06% ($Z = 1.379$, $p = 0.168$) - a clear pass.

**[F9] Digit mean and variance:** Mean = 4.557; variance = 8.278. Both consistent with the uniform distribution.

**[F10] Klimek mixture model:** The BIC-selected five-component model (BIC = −10,826.1) identifies a swing cluster (41.0%; turnout 64.0%, Democratic share 54.0%), Yeongnam I (27.6%; 66.1%, 42.5%), Yeongnam II (14.6%; 66.1%, 24.3%), Honam I (8.7%; 65.9%, 68.9%), and Honam II (8.2%; 66.2%, 86.5%). No cluster meets the Klimek fraud criterion.

**[F11] Sebastopol test:** One constituency flags: Sejong ($r = 0.726$, $n = 11$ Dongs), consistent with the structural explanation advanced in Section 3.5.

## 3.11 20th Presidential Election (2022): Nationwide Digit Tests [F1] and [F2]

**2BL [F1]:** Applied to the full $N = 3{,}509$ Dong-level `in_precinct_early_dem` totals, the chi-square test yields $\chi^2 = 13.87$ ($p = 0.127$, $df = 9$). We fail to reject the null at the $\alpha = 0.05$ threshold; second digits conform to the natural Benford distribution.

**Last-Digit Uniformity [F2]:** This is the most significant forensic result across all four elections. The test reveals an *asymmetric* failure pattern: only the in-precinct early Democratic vote series (`in_precinct_early_dem`) fails ($\chi^2 = 24.98$, $p = 0.003$, Cramér's $V = 0.028$), while the in-precinct early conservative series (`in_precinct_early_con`: $\chi^2 = 9.43$, $p = 0.399$) and the same-day Democratic series (`same_day_dem`: $\chi^2 = 8.42$, $p = 0.492$) both pass comfortably. The observed deviation in the Democratic early series exceeds what pure binomial sampling predicts from the empirical ($\texttt{in_precinct_early_total}_i, p_{\text{dem},i}$) distribution. Unlike the marginal 2BL results in the legislative elections, this result cannot be attributed to distributional homogeneity effects because the last-digit test is substan-

tially less sensitive to such conditions than the 2BL. Visual inspection of the last-digit bar chart (Figure 3, row 3, left panel) confirms a pronounced excess frequency at digit 5 relative to the expected uniform distribution. The diagnostic explicitly flags this asymmetry: the failure *'cannot be explained by a counting machine artifact (which would affect all candidates equally). It is candidate-specific.'* The failure of `in_precinct_early_dem` while `in_precinct_early_con` passes on the exact same ballots, counted by the same machines, at the same stations, constitutes the principal forensic finding of this study.

### 3.12 20th Presidential Election (2022): Variance Dispersion [F3], Covariate Standardisation, and PSM

Mean within-city vote-share standard deviation: 3.96% - the lowest across all four elections, reflecting the close and relatively homogeneous partisan landscape of the 2022 presidential contest.

The raw mean early-to-same-day gap is +7.04% ($\sigma = 3.72\%$) at Dong level and +6.20% ($\sigma = 2.66\%$) at constituency level. These are the smallest gap magnitudes of the four elections, consistent with the Democratic candidate's narrow overall performance advantage in early voting in a cycle where the conservative candidate prevailed. The WLS mobilisation model [C1] explains 54.8% of gap variance. The bivariate $r^2$ between `same_day_dem` share and `early_pct` is 0.9694 (Dong) and 0.9835 (constituency) - the highest across all four elections, implying the tightest linear relationship between early and same-day Democratic shares. The full WLS [C2] model achieves $R^2 = 0.983$.

Visual inspection of the Q-Q plot for the WLS residuals confirms that departures from normality are concentrated in the distribution tails rather than in the central mass. The absentee vote shift is $\mu = -0.40\%$ ($\sigma = 0.92\%$). PSM yields ATT = +7.05 pp.

## 3.13 20th Presidential Election (2022): Regional Aggregates [F5/F6] and Invalid Votes [F4]

**Overall two-party ratios [F6]:** Democratic (Lee) two-party shares: Gyeonggi 52.57% ($n$ = 3,825), Incheon 50.76% ($n$ = 885), Seoul 46.83% ($n$ = 2,689). These reflect the competitive nature of the election (the Democratic candidate narrowly lost nationally) and confirm Gyeonggi as a Democratic stronghold even in a losing cycle, while Seoul's 46.83% is consistent with Lee's underperformance in the capital. Micro standard deviation across metropolitan stations: 8.87%.

**Invalid vote correlation [F4]:** $r = -0.105$ ($p < 0.001$) across $N$ = 17,974 stations. This is the strongest invalid-vote correlation observed across all four elections, indicating a meaningful negative association between invalid-ballot rates and Democratic vote share. The diagnostic flags this correlation as warranting further investigation, though a negative sign direction does not mechanically imply fraud: invalid ballot rates could plausibly be higher in districts with older, less party-affiliated voter pools that also lean conservative.

## 3.14 20th Presidential Election (2022): Metro Early-Voting Analysis [F7]

**Metro 2BL [F7a]:** For the metro precincts, the native $\chi^2$ statistic is 8.85 ($p = 0.451$, $df = 9$) - a comfortable pass. The Shikano and Mack (2011) homogeneity diagnostic shows a metro Dong-level mode of approximately 2,278 votes and a KDE maximum of approximately 0.00026, placing the data clearly outside the S&M calibration range.

**Simulation-adjusted metro 2BL [F7d]:** When the 2BL is applied per constituency to the metropolitan subset, only 1 of 77 constituencies (1.3%) rejects at the 5% threshold, against an expected rejection rate of approximately 5% under the null - the lowest rejection rate across all four elections.

**Metro last-digit [F7b]:** $\chi^2 = 10.30$ ($p = 0.327$, $df = 9$). A clear pass on the metro subset, directly contradicting metro-level digit manipulation.

**Early-only two-party ratio [F7c]:** The macro in-precinct early-voting two-party

Democratic shares are: Gyeonggi 57.00% vs 43.00% ($n = 560$), Incheon 54.54% vs 45.46% ($n = 155$), Seoul 53.65% vs 46.35% ($n = 425$). These ratios are consistent with a Democratic early-voting premium of approximately 6–7 percentage points above same-day performance across all three zones, tracking the raw gap magnitude of +7.04% and reflecting the same age-based mobilisation differential as in the legislative elections. The 63:37 early-voting ratio does not arise in the 20th Presidential; Democratic early-voting shares range from 53% to 57%, substantially below the 63% legislative-election figure and proportional to the Democratic candidate's lower overall performance in 2022. The micro-level standard deviation of early-station two-party shares is 7.03%, confirming natural dispersion inconsistent with a top-down algorithmic target.

**Interpretation of the last-digit failure.** The nationwide last-digit result ($\chi^2 = 24.98$, $p = 0.003$) is the most significant forensic anomaly identified across all four elections and cannot be dismissed by appeal to distributional homogeneity effects or large-$N$ artefacts. Several observations structure its interpretation.

*First, the failure is asymmetric and candidate-specific.* The anomalous digit distribution affects only `in_precinct_early_dem`. The `in_precinct_early_con` series ($p = 0.399$) and `same_day_dem` series ($p = 0.492$) pass on the same ballots, at the same stations, processed through the same tabulation infrastructure. Unlike a cross-party failure - which could plausibly arise from a counting-machine or recording artefact affecting all vote types equally - this candidate-specific pattern is more difficult to attribute to a non-partisan tabulation artefact. This observation heightens rather than mitigates the forensic concern, and underscores the need for further investigation, particularly into the data recording and reporting practices for Democratic early votes in the 2022 presidential cycle.

*Second, the metropolitan tests pass cleanly.* The metro last-digit [F7b] passes ($p = 0.327$) and the sim-adjusted metro 2BL [F7d] yields a rejection rate of only 1.3% - the lowest across all four elections. If the nationwide last-digit failure were driven by large-scale manipulation in the competitive metropolitan areas (where the election was most contested), the metro tests should be more anomalous, not less. This spatial localisation points toward a non-metropolitan or small-precinct phenomenon.

*Third, the demographic mobilisation model retains explanatory power.* The WLS [C1] model explains approximately 54.8% of gap variance, comparable to the legislative elections (49.0% in 2020; 51.4% in 2024), indicating that the gap structure is consistent with organic demographic mobilisation even in the presence of the digit anomaly.

*Fourth, the gap magnitude and $R^2$ are structurally consistent.* The bivariate $R^2$ between `same_day_dem` share and `early_pct` (0.9694 at Dong level, 0.9835 at constituency level) is the highest across all four elections, implying the tightest alignment between early and same-day voting patterns rather than a systematic divergence consistent with targeted manipulation.

Taken together, three partially mitigating considerations prevent a definitive fraud interpretation: the sim-adjusted metro 2BL passes with the lowest rejection rate across all four elections (1/77, 1.3%); the metro last-digit passes ($p = 0.327$); and the demographic and structural patterns are consistent with organic mobilisation. The asymmetric, candidate-specific last-digit failure nonetheless remains forensically unresolved and warrants further investigation - particularly regarding data recording practices for `in_precinct_early_dem` in the 2022 presidential cycle - as a priority for future election forensics research on South Korea.

## 3.15 20th Presidential Election (2022): Additional Forensic Tests [F8]–[F11]

**[F8] Beber and Scacco '0s and 5s':** Observed frequency 19.78% ($Z = -0.3292$, $p = 0.742$) - a comfortable pass, providing no evidence of human fabrication in the terminal-digit distribution.

**[F9] Digit mean and variance:** Mean = 4.475; variance = 8.495 (expected: 8.250). The slight variance elevation is not statistically significant and passes the test.

**[F10] Klimek mixture model:** The BIC-selected four-component model (BIC = −16,946.7) identifies a swing cluster (34.8%; turnout 73.9%, Democratic share 49.1%), Yeongnam I (29.1%; 74.9%, 37.2%), Honam (18.1%; 79.3%, 86.3%), and Yeongnam II (18.0%; 77.6%, 21.0%). Presidential turnout levels are uniformly higher across all clusters

than in the legislative elections, consistent with the competitive 2022 cycle. No cluster meets the Klimek joint fraud criterion.

**[F11] Sebastopol test:** Three constituencies flag: Sejong ($r = 0.815$, $n = 22$), Heungduk ($r = 0.748$, $n = 11$), and Masan-Habpo ($r = 0.723$, $n = 15$). Sejong's recurrence across all four elections supports the structural explanation. In the context of the last-digit anomaly identified in [F2], the Heungduk and Masan-Habpo flags warrant noting; however, high turnout-share correlations can arise from strong within-constituency geographic partisan gradients without manipulation, and the Klimek test [F10] finds no corresponding nationwide fraud cluster, constraining any manipulation inference.

## 3.16 21st Presidential Election (2025): Nationwide Digit Tests [F1] and [F2]

**2BL [F1]:** $\chi^2 = 8.53$ ($p = 0.481$, $df = 9$). Nationwide second digits of `in_precinct_early_dem` conform closely to the Benford distribution. The nationwide distribution spans 31–13,448 votes (2.64 orders of magnitude).

**Last-Digit Uniformity [F2]:** All three series pass: `in_precinct_early_dem` ($\chi^2 = 11.09$, $p = 0.270$), `in_precinct_early_con` ($\chi^2 = 10.49$, $p = 0.311$), and `same_day_dem` ($\chi^2 = 5.00$, $p = 0.834$). No algorithmic padding signature. The clean, symmetric pass across all three series - in sharp contrast to the asymmetric failure of the 20th Presidential - confirms that the 2022 last-digit anomaly was specific to that cycle and does not reflect a structural feature of South Korean presidential election data.

## 3.17 21st Presidential Election (2025): Variance Dispersion [F3], Covariate Standardisation, and PSM

Mean within-city vote-share standard deviation: 4.00%.

The raw mean early-to-same-day gap is +22.47% ($\sigma = 6.75\%$) at Dong level and +20.79% ($\sigma = 5.58\%$) at constituency level. These are by far the largest gap magnitudes of the four elections, exceeding the legislative-election gaps by approximately 13

percentage points. This large differential reflects two reinforcing factors: the Democratic candidate's decisive overall margin in 2025 (approximately 61% nationally), which implies a larger absolute partisan differential in early versus same-day composition; and the unprecedented political salience of the election (the first since Yoon's impeachment), which may have differentially mobilised middle-aged early voters toward the opposition. The WLS mobilisation model [C1] explains 73.3% of gap variance (adj. $R^2 = 0.730$, $N = 3{,}553$). The bivariate $r^2$ between `same_day_dem` share and `early_pct` is 0.8972 (Dong) and 0.9302 (constituency) - lower than all other elections. The WLS [C2] model achieves $R^2 = 0.961$.

The Q-Q plots of the WLS residuals confirm that departures from normality remain concentrated in the tails, consistent with the other electoral cycles. The absentee vote shift is $\mu = -1.37\%$ ($\sigma = 0.89\%$).

## 3.18 21st Presidential Election (2025): Regional Aggregates [F5/F6] and Invalid Votes [F4]

**Overall two-party ratios [F6]:** Democratic (Lee) two-party shares across all metro stations: Gyeonggi 55.97% ($n = 3{,}888$), Incheon 55.47% ($n = 898$), Seoul 50.49% ($n = 2{,}686$). Micro standard deviation across metropolitan stations: 13.33% - the highest across all four elections, consistent with greater geographic polarisation in the presidential race.[1]

**Invalid vote correlation [F4]:** $r = -0.049$ ($p < 0.001$) across $N = 17{,}844$ stations - small, in the same direction as the 20th Presidential, and consistent with the legislative elections.

[1] These [F6] overall ratios are substantially lower than the early-only [F7c] ratios of 71–74%, as expected given that [F6] aggregates same-day and early votes. The [F6] values are lower than might be expected given the Democratic national landslide; a potential data-sourcing issue affecting the presidential column normalisation for [F6] cannot be excluded and warrants verification against raw NEC tabulations. Cross-election comparisons relying on [F6] for the 21st Presidential should be treated with caution.

## 3.19 21st Presidential Election (2025): Metro Early-Voting Analysis [F7]

**Metro 2BL [F7a]:** The Shikano and Mack (2011) diagnostic for the 21st Presidential metro subset shows a mode of approximately 2,750 votes and a KDE maximum of approximately 0.00023, placing the data clearly outside the S&M calibration range. The native chi-square statistic yields $\chi^2 = 20.40$ ($p = 0.016$, $df = 9$), failing at the 5% threshold. However, this must be evaluated alongside the simulation-adjusted test.

**Simulation-adjusted metro 2BL [F7d]:** 4 of 80 constituencies (5.0%) reject at the 5% threshold - exactly at the expected null rejection rate. A 5.0% rejection rate is precisely what one expects under the null hypothesis of natural digit processes across 80 independent tests.

**Metro last-digit [F7b]:** $\chi^2 = 7.24$ ($p = 0.613$, $df = 9$). Passes clearly, consistent with the nationwide [F2] pass.

**Early-only two-party ratio [F7c]:** The macro in-precinct early-voting two-party Democratic shares are: Gyeonggi 73.66% vs 26.34% ($n = 601$), Incheon 72.19% vs 27.81% ($n = 156$), Seoul 71.98% vs 28.02% ($n = 426$). These ratios are substantially higher than in all three prior elections, consistent with the Democratic candidate's large overall margin. The micro-level standard deviation is 6.35%, confirming natural dispersion. The 63:37 ratio does not arise; Democratic early-voting metro shares range from 71% to 74%. The dynamic progression across elections - 63% in 2020, 53–57% in 2022, 59–62% in 2024, 71–74% in 2025 - tracks Democratic overall performance monotonically, providing the strongest cross-election evidence against any fixed algorithmic target.

## 3.20 21st Presidential Election (2025): Additional Forensic Tests [F8]–[F11]

**[F8] Beber and Scacco '0s and 5s':** Observed frequency 20.83% ($Z = 1.233$, $p = 0.218$) - a clear pass.

**[F9] Digit mean and variance:** Mean = 4.558; variance = 8.205. Both consistent

with the uniform distribution.

**[F10] Klimek mixture model:** The BIC-selected four-component model (BIC = −17,726.0) identifies a swing cluster (37.6%; turnout 76.4%, Democratic share 49.5%), Yeongnam I (27.6%; 76.1%, 38.0%), Yeongnam II (17.0%; 78.0%, 21.8%) and Honam (17.8%; 81.0%, 86.6%). No cluster meets the Klimek fraud criterion. Turnout levels are uniformly high across all clusters, reflecting the elevated participation in the snap presidential election.

**[F11] Sebastopol test:** Six constituencies are listed as flagging: Sejong ($r = 0.876$, $n = 24$), Heungduk ($r = 0.819$, $n = 11$), Yangyang ($r = 0.815$, $n = 6$), Jeju ($r = 0.791$, $n = 26$), Dalseo ($r = 0.722$, $n = 23$) and Masan-Habpo ($r = 0.768$, $n = 15$). This is the highest flagged count across all four elections. Critically, Sejong, Heungduk, and Masan-Habpo recur across multiple elections (Sejong in all four, Heungduk and Masan-Habpo in both presidential cycles), again strongly supporting a structural constituency-level explanation. The Yangyang flag arises in a very small constituency ($n = 6$ Dongs), where the correlation statistic is mechanically sensitive to individual outlier Dongs. The Jeju flag is more notable given the larger $n = 26$; however, the Klimek test [F10] finds no corresponding nationwide fraud cluster, and Jeju's geographic isolation may generate a natural within-constituency turnout-partisanship gradient.

## 3.21 Cross-Election Comparison

Table 1 summarises all key statistics across the four elections.

Several observations emerge from the cross-election comparison. First, all five nationwide tests [F1]–[F5/F6] pass in both legislative elections with similar chi-square statistics, confirming stable natural digit properties. Second, the 20th Presidential last-digit failure ($\chi^2 = 24.98$, `in_precinct_early_dem` only) stands qualitatively apart from any result in the other three elections: it exceeds the next-highest nationwide digit statistic (the 20th Presidential 2BL at 13.87) and the marginal 22nd General metro 2BL (18.00) in a test that is less susceptible to structural confounds, and - critically - it is asymmetric, affecting only the Democratic early series while `in_precinct_early_con` passes on the same bal-

Table 1: Key forensic statistics across all four elections (2020–2025)

| Metric | 21st Gen (2020) | 20th Pres (2022) | 22nd Gen (2024) | 21st Pres (2025) |
|---|---|---|---|---|
| *Nationwide digit and distribution tests* | | | | |
| 2BL $\chi^2$ ($p$) [F1] | 9.36 (0.404) | 13.87 (0.127) | 10.51 (0.311) | 8.53 (0.481) |
| Last-digit $\chi^2$ ($p$) [F2], `in_precinct_early_dem` | 6.22 (0.718) | 24.98 (0.003)$^{\ddagger}$ | 11.05 (0.272) | 11.09 (0.270) |
| Last-digit $\chi^2$ ($p$) [F2], `in_precinct_early_con` | 9.59 (0.385) | 9.43 (0.399) | 6.65 (0.674) | 10.49 (0.312) |
| Last-digit $\chi^2$ ($p$) [F2], `same_day_dem` | 9.53 (0.390) | 8.42 (0.492) | 7.43 (0.593) | 5.00 (0.834) |
| Mean within-city SD [F3] | 5.85% | 3.96% | 5.25% | 4.00% |
| Invalid vote $r$ [F4] | −0.039 | −0.105$^{*}$ | +0.081 | −0.049 |
| *Metro early-voting analysis [F7]* | | | | |
| Metro 2BL $\chi^2$ ($p$) [F7a] | 16.38 (0.059) | 8.85 (0.451) | 18.00 (0.035, fail$^{\dagger}$) | 20.40 (0.016, fail$^{\dagger}$) |
| Metro last-digit $\chi^2$ ($p$) [F7b] | 13.28 (0.150) | 10.30 (0.327) | 12.17 (0.204) | 7.24 (0.613) |
| Sim-adj metro 2BL [F7d] | 9/114 (7.9%) | 1/77 (1.3%, pass) | 5/119 (4.2%) | 4/80 (5.0%) |
| Early in-precinct dem share, Gyeonggi [F7c] | 63.27% | 57.00% | 61.78% | 73.66% |
| Early in-precinct dem share, Incheon [F7c] | 63.07% | 54.54% | 59.02% | 72.19% |
| Early in-precinct dem share, Seoul [F7c] | 63.48% | 53.65% | 59.61% | 71.98% |
| Metro early micro SD [F7c] | 9.14% | 7.03% | 7.04% | 6.35% |
| *Additional forensic tests [F8]–[F11]* | | | | |
| 0s and 5s $p$-value [F8] | 0.426 | 0.742 | 0.168 | 0.218 |
| Klimek fraud cluster [F10] | None | None | None | None |
| Sebastopol flags ($r > 0.70$) [F11] | 1 (Sejong) | 3 (Sejong,Heungduk,Masan-Habpo) | 1 (Sejong) | 6 (Yangyang,Jeju,Dalseo,+)$^{\S}$ |
| *Gap magnitude* | | | | |
| Raw gap, Dong ($\mu$, $\sigma$) | +8.89%, 4.88% | +7.04%, 3.72% | +9.60%, 5.02% | +22.47%, 6.75% |
| Raw gap, constituency ($\mu$, $\sigma$) | +10.29%, 3.00% | +6.20%, 2.66% | +10.37%, 2.96% | +20.79%, 5.58% |
| Absentee shift ($\mu$, $\sigma$) | +0.50%, 0.76% | −0.40%, 0.92% | +0.52%, 0.68% | −1.37%, 0.89% |
| *Regression and matching* | | | | |
| WLS gap adj. $R^2$ [C1] | 0.484 | 0.542 | 0.507 | 0.730 |
| Bivariate $r^2$, Dong | 0.9186 | 0.9694 | 0.9464 | 0.8972 |
| WLS ratio $R^2$ (covariates) [C2] | 0.961 | 0.983 | 0.968 | 0.961 |
| PSM ATT [C3] | +4.53 pp | +7.05 pp | +2.09 pp | +4.65 pp |

$^{\ddagger}$ Asymmetric failure: `in_precinct_early_dem` fails ($\chi^2 = 24.98$, $p = 0.003$) while `in_precinct_early_con` ($p = 0.399$) and `same_day_dem` ($p = 0.492$) both pass on the same ballots and stations. Candidate-specific; cannot be attributed to a counting machine artefact. Warrants further investigation.

Metro last-digit passes ($p = 0.327$); sim-adj metro 2BL passes (1/77 = 1.3%, lowest across all four elections).

$^{*}$ Warrants further investigation; see text.

$^{\dagger}$ Native test failures (22nd Gen: $\chi^2 = 18.00$; 21st Pres: $\chi^2 = 20.40$). Both pass the sim-adj [F7d] constituency test (4.2% and 5.0% respectively), arguing against fraud.

$^{\S}$ as well as Sejong, Heungduk, Masan-Habpo. F9 (digit mean and variance) passes in all four elections.

lots. Third, the 21st Presidential passes all digit and structural tests symmetrically across all three series, confirming that the 2022 last-digit anomaly was cycle-specific rather than structural to presidential elections. Fourth, the simulation-adjusted metro 2BL [F7d] passes in all four elections: 9/114 (7.9%, borderline) for the 21st General, 5/119 (4.2%) for the 22nd General, 1/77 (1.3%) for the 20th Presidential, and 4/80 (5.0%) for the 21st Presidential. Fifth, the metro early-voting Democratic shares progress monotonically with Democratic overall performance across all four elections (63%, 53–57%, 59–62%, 71–74%), providing powerful evidence against any fixed algorithmic target. Sixth, the absentee shift is structurally positive and similar in both legislative elections, reverses sign in both presidential elections, suggesting that the absentee voter pool composition differs systematically between election types. Seventh, all elections pass [F8] (0s and 5s), [F9] (digit mean/variance), and [F10] (Klimek) across all four cycles. The [F11] Sebastopol test flags Sejong in all four elections, Heungduk and Masan-Habpo in both

presidential elections, consistent with structural constituency-level explanations. Finally, the WLS mobilisation models explain 49–73% of gap variance across all four elections through demographic covariates, confirming that the dominant driver of the early-voting gap is demographic mobilisation in every cycle examined.

# 4 Conclusion

Our comprehensive frequentist analysis finds no statistical evidence of artificial data generation in either the 21st (2020) or the 22nd (2024) South Korean legislative elections, nor in the 21st Presidential Election (2025). All nationwide digit-based and structural forensic tests pass in these three cycles. The 63:37 metropolitan early-voting ratio is confirmed for the 21st General Election but ranges from 53% to 74% across the other three elections, dynamically tracking the Democratic candidate's overall performance - behaviour that is incompatible with a fixed algorithmic target across any of the four cycles. Human-fabrication digit tests [F8]/[F9], the Klimek mixture-model fingerprint [F10], and the Sebastopol constituency-level correlation test [F11] return no fraud signals in any of the four elections, subject to the structural interpretation of Sejong's persistent Sebastopol flag.

The failures of the metro-restricted 2BL in the 22nd General ($\chi^2 = 18.00$, $p = 0.035$) and 21st Presidential ($\chi^2 = 20.40$, $p = 0.016$) elections do not constitute evidence of fraud. While the Shikano and Mack (2011) diagnostic confirms that the station-level inflation mechanism does not operate at Dong-level aggregation, both elections cleanly pass the simulation-adjusted constituency-level test (rejecting at 4.2% and exactly 5.0%, respectively). Furthermore, their nationwide [F1] tests pass cleanly, and the metro last-digit [F7b] tests pass on the exact same data.

The most significant finding is a genuine nationwide last-digit test failure in the 20th Presidential Election (2022): $\chi^2 = 24.98$ ($p = 0.003$), confined exclusively to the `in_precinct_early_dem` series, while `in_precinct_early_con` ($p = 0.399$) and `same_day_dem` ($p = 0.492$) pass on the same ballots and stations. This asymmetric, candidate-specific

failure exceeds the critical value by 7.90 - far more than the marginal legislative 2BL results - and is not replicated in the 21st Presidential Election (2025). Three partially mitigating considerations prevent a definitive fraud interpretation: the simulation-adjusted metro 2BL yields the lowest rejection rate across all four elections (1/77, 1.3%); the metro last-digit passes ($p = 0.327$); and demographic covariates explain 54.8% of gap variance while the bivariate $R^2$ between same-day and early shares (0.9694 at Dong level) is the highest of all four elections, implying tighter alignment rather than targeted divergence. The asymmetric, candidate-specific nature of the failure - confined to `in_precinct_early_dem` while `in_precinct_early_con` and `same_day_dem` pass on the same infrastructure - cannot however be explained by a generic tabulation artefact and represents the most forensically unresolved result across all four elections examined. It warrants targeted investigation of data recording and reporting practices for the 2022 presidential `in_precinct_early_dem` series as a priority for future research.

Several broader methodological lessons emerge from this four-election analysis. First, Bayesian mixture models must incorporate polarising demographic covariates or they will systematically misclassify natural sociological sorting as artificial manipulation; this lesson applies with equal force to presidential elections where partisan polarisation is at its greatest. Second, digit-based tests should be interpreted jointly and disaggregated by vote series: the 20th Presidential [F2] failure, which manifests asymmetrically in `in_precinct_early_dem` while `in_precinct_early_con` passes, carries substantially different inferential content than a symmetric cross-party failure would. Third, the last-digit test is more sensitive to candidate-specific anomalies than the 2BL, and disaggregating by series is essential to distinguish artefacts from manipulation candidates. Fourth, the monotonic relationship between Democratic overall performance and metro early-voting shares across all four elections ($r \approx 1.0$ at the zone level) is perhaps the most powerful single piece of evidence against a fixed-target algorithmic scheme, precisely because it holds across elections with very different political contexts and electoral rules.

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# Appendix A: Code Repository

The Python code and the resulting data logs used to generate this research is located at `https://github.com/timothyalphakim/southkorea_election/`. The code and manuscript will be updated regularly there.